

Exam Date & Time: 27-Jun-2024 (02:30 PM - 05:30 PM)



MANIPAL ACADEMY OF HIGHER EDUCATION

MIT BLR - BTech VI Semester - End Semester Makeup Examination - June -2024

Agile Software Development for Engineers [IT-4308]

Marks: 50

Duration: 180 mins.

PART-A

Answer all the questions.

- 1) How does Agile development promote adaptability and responsiveness to change compared to traditional software development methodologies? Provide specific examples from Agile practices. (4)
 - A)
 - B) Illustrate the steps used to transfer from traditional approach to agile process by considering Education Sector for Curriculum Development and Adaptive Learning. (3)
 - C) Give the need of DevOps. Describe the Responsibilities, and Skills that is expected from DevOps engineer (3)
- 2) What is site reliability engineering? Discuss the benefits of site reliability engineering (SRE) practices. (4)
 - A)
 - B) In what ways does the Kanban system assist in managing and optimizing workflow within a team? Furthermore, what role do Work in Progress (WIP) limits play in this process? (3)
 - C) Illustrate the process of scrum framework. (3)
- 3) Illustrate how Agile teams achieve continuous delivery, and what benefits does it offer in terms of speed, quality, and customer satisfaction? (4)
 - A)
 - B) In what ways can virtual meetings enhance the evaluation of Agile methodologies in distributed startup teams? (3)
 - C) How does Agile handle requirements that change frequently during a project? (3)
- 4) Illustrate how organizations combine Kanban and Agile practices to achieve a flexible and adaptive approach to project and product development. (4)
 - A)
 - B) How does Agile address quality assurance and testing practices? (3)

- C) What distinguishes self-organizing teams from traditional hierarchical structures, and how does it promote innovation and productivity? (3)
- 5) How does the agile principle "Working software over comprehensive documentation" contribute to project success? (4)
 - A)
 - B) Outline the potential challenges associated with implementing Test-Driven Development (TDD) in a large, existing codebase? How can these challenges be mitigated? (3)
 - C) Discuss the Agile practice of "Pair Programming." What benefits does it offer, and how does it contribute to the quality of the software product? (3)

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